

11 August 2026

Editor-in-Chief

Computers & Electrical Engineering

Dear Editor,

Please consider our Research Article, “ASTER: Authorization-Scoped Threshold Exact-Domain Credential Derivation with Failure-Safe Root-Epoch Healing,” for publication in *Computers & Electrical Engineering*.

The manuscript addresses a practical security problem in enterprise computing: legacy applications often retain password-only authentication even when stronger credential governance is required. ASTER combines exact accepted-policy-space sequencing, threshold evaluation without endpoint key reconstruction, generation-scoped authorization, and failure-safe Root-Epoch replacement after complete old-root disclosure. The contribution is the credential lifecycle contract connecting these mechanisms; the paper does not claim new secret-sharing, multiparty-computation, oblivious-PRF, format-preserving-encryption, or finite-language primitives.

The evaluation includes 9.7 million successful derivations across 97 supported exact policy domains, authorization-scope experiments, post-compromise migration experiments, 96 fault-injection traces, bounded TLA+ model checking, a real OpenLDAP interoperability smoke test, and malicious honest-majority MP-SPDZ fixed-vector executions. The reported MPC measurements are explicitly limited to feasibility and cost boundaries rather than production performance.

The work fits the journal’s security and privacy scope and emphasizes computational techniques with practical relevance to enterprise systems. The manuscript is original, has not been published previously, and is not under consideration by another journal.

Thank you for considering this submission.

Sincerely,

Yuanyi Zhang

Corresponding author

Hangzhou Information Technology Branch, Information Technology Institute, China Railway
Shanghai Group Co., Ltd.

revanton@icloud.com